

MATAD, Mongolia

WMO: 443140

Lat: 46.9550N

Lon: 115.2964E

Elev: 2976

StdP: 13.18

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-17.3	-27.7	1.3	-20.8	-23.8	1.6	-16.6	30.4	1.6	27.3	2.7	6.3	320	0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	90.4	63.7	86.3	62.2	82.8	61.4	67.7	83.5	65.7	78.9	64.1	76.3	10.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.5	98.0	72.0	61.7	91.9	70.1	59.9	85.9	69.2	34.0	83.5	32.3	79.1	31.0	75.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	23.6	22.1	-27.4	95.5	3.8	5.2	-30.1	99.3	-32.3	102.3	-34.4	105.2	-37.1	109.0		
(5)			-27.4	71.2	4.2	2.5	-30.4	73.0	-32.9	74.5	-35.2	75.9	-38.3	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	-2.0	5.2	19.7	38.2	52.8	64.3	69.8	65.6	54.2	36.4	15.5	2.2		
	DBStd	26.83	8.56	10.27	11.39	9.70	8.67	7.49	5.63	6.48	8.59	9.61	11.79	8.61		
	HDD50	7252	1613	1254	940	370	68	1	0	1	53	434	1036	1482		
	HDD65	11191	2078	1674	1404	803	389	101	14	68	339	888	1486	1947		
	CDD50	1890	0	0	0	18	155	431	613	484	178	11	0	0		
	CDD65	356	0	0	0	0	12	80	163	87	14	0	0	0		
	CDH74	4078	0	0	0	9	253	1029	1626	960	199	2	0	0		
	CDH80	1412	0	0	0	1	77	374	595	319	46	0	0	0		
	WSAvg	9.7	9.5	9.7	10.2	11.6	11.7	9.2	8.2	8.1	9.2	9.8	9.4	10.2		
	PrecAvg	11.8	0.1	0.1	0.3	0.5	1.0	2.2	3.3	2.3	1.0	0.4	0.3	0.2		
	PrecMax	22.2	0.4	0.3	0.5	1.4	2.9	4.1	9.5	5.4	2.2	1.0	0.9	0.5		
(15) Precipitation	PrecMin	6.5	0.0	0.0	0.0	0.1	0.2	0.2	0.9	0.9	0.2	0.1	0.0	0.1		
	PrecStd	3.3	0.1	0.1	0.1	0.3	0.6	0.9	1.9	1.1	0.5	0.2	0.2	0.1		
	DB	24.0	37.8	57.6	73.8	88.1	94.9	96.4	93.9	85.5	70.0	49.0	29.0			
	MCWB	20.2	29.9	40.4	49.5	56.2	64.0	67.6	67.7	60.1	50.4	37.1	24.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	18.7	30.6	49.4	67.0	80.7	88.3	90.9	87.2	80.2	62.8	42.7	22.5			
	2% MCWB	15.7	25.0	35.9	46.0	53.9	60.9	64.8	64.4	58.1	46.7	33.7	19.4			
	5% DB	14.5	25.6	42.9	61.1	75.4	84.2	86.5	83.0	75.0	56.7	37.3	18.3			
	5% MCWB	12.2	21.4	32.3	42.9	52.7	59.8	64.3	62.4	56.1	42.7	30.0	15.8			
	10% DB	10.7	21.4	37.5	56.4	70.7	79.9	82.8	79.2	70.1	52.0	32.6	14.8			
	10% MCWB	8.8	17.9	29.2	40.6	50.0	59.1	63.3	60.9	53.8	40.3	27.1	12.7			
	0.4% WB	20.9	30.8	40.8	50.8	59.1	67.5	71.2	70.7	64.0	52.1	38.3	25.8			
	0.4% MCDB	23.8	36.7	55.8	72.1	79.3	87.5	86.7	87.8	77.3	68.1	48.8	30.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	16.2	25.7	36.5	46.9	56.1	63.9	68.4	66.6	59.9	47.1	34.6	20.1			
	2% MCDB	19.1	31.2	49.1	65.0	75.5	80.6	83.9	80.2	74.2	61.2	43.9	23.3			
	5% WB	12.9	22.1	33.1	43.7	53.7	62.3	66.6	64.6	57.5	44.0	31.5	16.8			
	5% MCDB	15.3	26.3	43.1	59.9	72.6	77.6	80.6	76.7	71.1	55.4	38.9	19.3			
	10% WB	9.4	18.9	29.9	40.8	51.2	60.7	65.1	63.1	55.2	40.9	28.6	13.8			
	10% MCDB	11.1	22.4	37.8	55.7	68.9	75.4	77.4	74.6	68.1	51.0	34.1	15.9			
	MDBR	16.5	18.9	21.2	23.4	24.7	21.9	20.2	21.6	23.1	21.3	18.3	15.1			
	5% DB	20.9	23.1	27.7	31.1	31.9	28.1	26.2	27.6	29.0	27.7	23.4	17.5			
(36) Mean Daily Temperature Range	5% MCWBR	18.0	18.4	17.8	17.5	15.4	11.5	9.9	11.7	14.2	16.3	15.8	14.6			
	5% WB	20.1	22.7	27.0	28.6	28.0	23.5	21.7	22.0	26.0	26.3	22.6	17.3			
	5% MCWBR	17.9	18.3	17.7	16.9	15.0	10.5	9.7	10.3	13.9	16.1	15.6	14.8			
	taub	0.253	0.274	0.315	0.373	0.442	0.454	0.415	0.377	0.336	0.307	0.273	0.257			
(40) Clear-Sky Solar Irradiance	taud	2.322	2.291	2.255	2.144	1.982	2.066	2.254	2.328	2.381	2.385	2.355	2.297			
	Ebn at Noon	269	285	287	278	261	259	268	275	278	269	258	251			
	Edh at Noon	23	30	36	45	55	51	42	37	32	27	22	21			
	RadAvg	652	1026	1472	1819	2050	2047	1958	1823	1477	1038	669	512			
(44) All-Sky Solar Radiation	RadStd	40	59	60	82	108	144	90	98	80	61	34	33			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-3.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page